

Background: FARS stands for the Fatality Analysis Reporting System. It is a database maintained by the National Highway Traffic Safety Administration (NHTSA), an agency of the U.S. federal government, and it records every fatal motor vehicle crash on public roads in the United States. It has run continuously since 1975 and covers all 50 states, the District of Columbia, and Puerto Rico. It is free, public, and downloadable without any application or approval process. A crash enters FARS only if all of the following hold: it involved a motor vehicle, it occurred on a road open to the public, and at least one person died within 30 days as a result.

FARS is not a sample of crashes. It is a complete count, called a census, of one specific type of crash. Crashes in which everyone survived are not included in the file at all. Neither are crashes on private property, nor are deaths that occur 40 days after the crash.

The practical consequence for modeling is that the dataset is selected on its outcome. You cannot ask FARS whether a crash was likely to happen, or whether a crash was likely to be fatal, because there is nothing in the file to compare against: every row is already a fatal crash. Any question you pose must be conditional on a fatal crash having occurred. This is fundamental to both the following proposals.

How the data is organized

FARS is not a single spreadsheet. It is a set of linked tables at three levels of detail.

One table is called ACCIDENT/CRASH, with one row per crash, with details like Date, time, location, weather, light conditions, road type, and how the vehicles collided) 2nd is the VEHICLE table, with one row per vehicle involved in that crash (body type, model year, speed, maneuver, damage, point of first impact), and third is the PERSON table, one row per person in or struck by that vehicle with details such as Age, sex, seat position, seat belt use, airbag, injury outcome, alcohol test result. For example, suppose a sedan carrying two people collides with a pickup truck carrying one person, and one of the sedan occupants dies. That single event produces:

one row in the crash table,

two rows on the vehicle table (the sedan and the pickup),

three rows on the person table (two sedan occupants, one pickup occupant).

All of these rows carry the same crash identifier, and the vehicle and person rows carry additional numbers identifying which vehicle and which person is within the crash. Joining the tables on these keys is how to build an analysis-ready table. Beyond the three main tables, FARS includes several supplementary tables covering specific topics, such as driver distraction, driver impairment, the sequence of events during the crash, and a dedicated table describing crash circumstances for pedestrians and cyclists.

FARS can be downloaded directly from NHTSA as annual files or acquired programmatically. It comes with a codebook table documenting every variable, its permitted values, their meanings, and which years each variable is valid for. That codebook is the authority for resolving the two problems described above.

Vocabulary used later in this document

Term	Meaning
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Occupant	A person inside a vehicle, whether driver or passenger.
Non-motorist	A person not in a vehicle: a pedestrian, a cyclist, or similar.
Restraint use	Whether a seat belt or child safety seat was being used.
Manner of collision	How the vehicles struck each other: head-on, rear-end, side impact, and so on.
Point of impact	Where on the vehicle the first contact occurred, recorded as a clock position.
Functional class	The official category of the road: interstate, arterial, local street, and so on.
Land use	Whether the location is classified as urban or rural.
Injury severity	The coded outcome for a person, ranging from no injury through to fatal injury.
BAC	Blood alcohol concentration. In the United States, 0.08 g/dL is the legal limit for driving in most circumstances.
Census (of crashes)	A complete count of every qualifying event, as opposed to a sample.
Exposure	A measure of how much driving took place, used as a denominator to convert crash counts into rates. FARS has no exposure measure of its own.

Proposal A: Who Survives a Fatal Crash?

1. Problem and Motivation: We are in Track 2, novel application. A fatal crash is defined by at least one death, but it does not follow that everyone involved dies. In a substantial share of fatal crashes, some occupants are killed while others in the same vehicle, or in the other vehicle, survive. FARS records an injury outcome for every person involved, not only for those who died. This asks the following question: **conditional on being involved in a crash severe enough to kill someone, what predicts whether a particular occupant survived?** The motivation is that this question isolates occupant protection at the extreme tail of crash severity. In an ordinary crash-severity study, the effect of a seat belt is entangled with the fact that belt wearers and non-wearers are involved in different kinds of crashes. Here, because the sample is restricted to crashes severe enough to result in a death, crash severity is held approximately fixed by construction, and what remains to explain differences in outcomes is occupant- and vehicle-level protection: restraint use, airbag deployment, seating position, occupant age, and vehicle design. These are precisely the factors that restraint campaigns, vehicle safety standards, and rear-seat occupant policies are meant to address.
2. Related Work and Positioning: Three streams of prior work are available on this question
 - Classical epidemiological analysis of FARS. This tradition estimates the effect of one factor at a time on whether a person dies, using conventional statistical models and reporting odds ratios rather than predictive performance. Its concern is explanation of a single variable, not prediction across a full feature set. Reference literature: The safest seat: effect of seating position on occupant mortality (DOI: [10.1016/j.jsr.2008.06.003](https://doi.org/10.1016/j.jsr.2008.06.003)) and Age-based variability in the association

between restraint use and injury type

(https://stacks.cdc.gov/view/cdc/124595/cdc_124595_DS1.pdf).

- Machine learning applied to crash injury severity. This strand does use predictive models, but it generally works on datasets that include non-fatal crashes and predicts a multi-level severity outcome, rather than asking who survives within crashes that were already fatal. Reference Literature: On Scene Injury Severity Prediction machine learning algorithms for motor vehicle crash occupants in the US (<https://doi.org/10.1016/j.jth.2021.101124>) and Crash injury severity prediction for passenger cars with machine learning algorithms (<https://doi.org/10.1093/eurpub/ckaf165.111>).
- And the third stream is about whether crash models transfer across states. This strand asks whether a model trained in some jurisdictions still performs in jurisdictions it never saw and finds that transfer learning helps. It has been established at the level of road segments and whether a crash occurs, not at the level of individual people and their outcomes, which is the gap your extension fills. Reference Literature: Graph Neural Networks for Road Safety Modeling (https://proceedings.neurips.cc/paper_files/paper/2023/file/a365be0950259c9624edfb4d26eabd46-Paper-Datasets_and_Benchmarks.pdf).

Neither strand treats survival in a fatal crash as a prediction problem and mostly separates features that exist before the crash from those that are its consequences. Our contribution has three parts. First, we take survival, conditional on a fatal crash having occurred, as the prediction target. Second, we divide the features into two blocks before modeling begins: those present before the crash, such as restraint use and vehicle design, and those that follow from it, such as ejection. We report a model for each, rather than including only consequence features. Third, we ask whether relationships learned in one set of states still hold in states the model never saw during training, a question neither strand addresses.

3. Approach and Data:

Data: FARS, obtained through NHTSA's public downloads (the annual files with flat tables containing one row per person with crash-level, vehicle-level, and person-level variables already merged, alongside a codebook table documenting variable names and value labels. We propose using calendar years 2016 through 2023 **(I already have the data with me)**

Unit of analysis: The person. We restrict to motor vehicle occupants and exclude non-motorists, because a pedestrian struck in a crash severe enough to enter FARS almost always dies, which would make the survival question degenerate for that group. This restriction should be stated openly as a scope decision.

Label: A binary indicator derived from the person-level injury severity variable: died of injuries sustained in this crash, versus survived. Records coded as having died prior to the crash, and records with unknown severity, are excluded rather than imputed.

Features- pre-crash and design: Restraint use, airbag presence and deployment, seating position, occupant age, occupant sex, vehicle body type, vehicle model year, manner of collision, initial impact point, light condition, weather, rural or urban land use, roadway functional class, and speeding involvement.

Features- crash consequence: Ejection, fire occurrence, and extent of vehicle damage. These are downstream of the crash and partly on the causal path to death, so a model that includes them will

predict well but explain little. We will fit the model both with and without this block and report the difference, which we expect to be one of the more interesting findings.

Baseline: Logistic regression on the pre-crash feature set, which is both the conventional choice in the epidemiological literature and directly interpretable.

Main models: Gradient-boosted trees, using XGBoost or LightGBM, with random forest as a secondary comparison. (Or we can decide to do something else)

Interpretation: SHAP values to identify which protective factors carry the most weight conditional on extreme crash severity.

Generalization extension: Train on a subset of states and evaluate on held-out states, and separately train on 2016 through 2020 and evaluate on 2021 through 2023. FARS is unusually well suited to this because every state codes its data to the same federal standard, which makes the state a clean way to define train and test domains. The same experiment run across separate state-run crash databases would be confounded, because each state defines and records its fields differently, so any performance drop could simply reflect those definitional differences rather than a real failure to generalize.